

Appendix C: IEEE 830 Template

Software Requirements Specification

for

Gym Management System

Version 0.3

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IFE – Software Engineering

18/03/2026

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Revision History

Jaime Esteban Reaño	04/03/2026	Initial version of the SRS document created. Project topic selected: Gym Management System.	0.1
Jaime Esteban Reaño	11/03/2026	Added Introduction and Overall Description sections up to 2.1 based on requirements analysis.	0.2
Jaime Esteban Reaño	18/03/2026	Updated product features and user classes according to UML analysis activities.	0.3

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1. Introduction

1.1. Purpose

This document specifies the software requirements for the Gym Management System, a software solution intended to support the daily management of a gym. The system will provide functionalities for member registration, class booking, trainer management, membership tracking, and payment control. This specification defines the current scope of the project and will be progressively updated as new analysis and design artifacts are developed during the course.

1.2. Document Conventions

Functional requirements are identified using the prefix FR, while non-functional requirements are identified using the prefix NFR. All mandatory requirements are written using the word “shall” to ensure precision and verifiability. UML diagrams included in the document are used as analysis and design support artifacts.

1.3. Intended Audience and Reading Suggestions

This document is intended for course instructors, project team members, testers, and future developers of the Gym Management System. Readers should begin with the Introduction and overall description sections to understand the purpose and scope of the project, and then continue with the system features and interface requirements

1.4. Product Scope

This Gym Management System aims to digitize and simplify the management of gym operations. Its main objective is to centralize information related to members, trainers, fitness classes, memberships, and payments in a single system. The expected benefits include improved administrative efficiency, reduced manual errors, easier scheduling, and better service for gym members

1.5. References

- Lab 1: Requirements Analysis – Software Engineering course materials
- Lab 2: UML Use Case Diagrams – Software Engineering course materials
- Lab 3: UML CLASS Diagrams – Software Engineering course materials
- IEEE 830 Software Requirements Specification Template, Appendix C course
- Lab 4: UML Sequence, State, and Activity Diagrams

2. Overall Description

2.1. Product Perspective

This Gym Management System is conceived as a new standalone software product. It is intended to replace informal or manual management methods based on spreadsheets, paper records, or disconnected applications. The system will centralize core gym operations in a single platform.

2.2. Product Features

- Member registration and profile management
- Membership plan management
- Trainer management
- Class scheduling and booking
- Payment tracking and membership validation
- Attendance monitoring

2.3. User Classes and Characteristics

The main user classes of the Gym Management System are Administrators, Trainers, and Members. Administrators are the primary management users and have the highest privilege level, being responsible for registering members, managing classes, assigning trainers, and controlling payments. Trainers use the system to consult assigned sessions, review schedules, and possibly track attendance. Members are end users who interact with the system to manage their personal information, check available classes, make reservations, and review membership status.

2.4. Operating Environment

This system is expected to operate as a web – based application. You could access it through modern browsers on desktop and also mobile devices. It should be compatible with common operating systems, like Windows, macOS, Android and even iOS , through browser access. However, an internet connection is required for its normal operation.

2.5. Design and Implementation Constraints

The project is developed in an academic environment and therefore its scope is limited to the core functionalities of a gym management platform. The system should prioritize clarity, usability, and modularity over advanced commercial features. Personal data related to members and trainers must be handled securely. The specification and design of the project will evolve incrementally according to the software engineering laboratory activities carried out during the course.

2.6. User Documentation

The system is expected to include basic user documentation adapted to an academic project context. This documentation may include a short administrator guide explaining how to register members, manage classes, assign trainers, and record payments, as well as a brief user guide for members describing how to view classes, make reservations, and review membership information. If implemented, online help or embedded interface hints may also be provided. The documentation will be delivered in digital format, most likely as PDF files and brief in-system textual guidance.

2.7. Assumptions and Dependencies

The specification assumes that the gym operates with clearly defined user roles, including administrators, trainers, and members. It is also assumed that the system will be accessed through devices with a stable internet connection and a compatible web browser. The project depends on decisions that may be taken in later stages of development, such as the database solution, the authentication mechanism, and the deployment environment. Since the project is developed incrementally in parallel with the laboratory activities of the course, some requirements and design decisions may evolve as new software engineering concepts are introduced.

3. System Features

3.1. Member Management

This feature covers the registration, storage, consultation, and update of member information. It also includes the management of membership status and related administrative data.

3.2 Trainer and Class Management

This feature supports the registration and management of trainers, as well as the creation and scheduling of fitness classes. It also includes the assignment of trainers to specific class sessions.

3.3 Booking and Attendance Management

This feature allows members to view available classes and book sessions when places are available. It also supports the recording or monitoring of attendance for scheduled activities.

4. External Interface Requirements

4.1. User Interfaces Overview

The Gym Management System will provide a graphical user interface (GUI) accessible via web browsers on desktop and mobile devices. Users will interact with the system through buttons, forms, and menus, entering data via text fields and selecting options from lists.

The system will offer different views depending on the user role (Administrator, Trainer, Member), allowing each user to perform their specific tasks such as managing data, viewing schedules, or booking classes.

The interface will provide clear feedback through confirmation and error messages, and will follow a consistent and user-friendly layout to ensure easy navigation and efficient use of the system.

4.2. Hardware Interfaces

The Gym Management System does not require any specific hardware interfaces, as it is designed to operate as a web-based application. Users will access the system through standard devices such as desktop computers, laptops, tablets, and smartphones.

The system interacts with hardware components indirectly through web browsers, using standard input and output devices such as keyboards, touchscreens, and displays. No specialized hardware or direct hardware communication protocols are required.

4.3. Software Interfaces

«Describe the connections between this product and other specific software components (name and version), including databases, interpreters, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications. Identify data that will be shared across software components. If the data sharing mechanism must be implemented in a specific way (for example, use of a global data area in a multitasking operating system), specify this as an implementation constraint.»

5. System Features/Modules

«This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product.»

5.1. «System Feature Name»

5.1.1 Description and Priority

«Provide a short description of the feature. Give its priority, too.»

5.1.2 Stimulus/Response Sequences

«"Stimulus" or "Condition"»: «Describe the stimulus or condition»

Response: «Describe the response»

«Additional Stimulus/Response pairs as needed»

5.1.3 Functional Requirements

«Itemize the detailed functional requirements associated with this feature. These are the software capabilities that must be present in order for the user to carry out the services provided by the feature. Include how the product should respond to anticipated error conditions or invalid inputs. Requirements should be concise, complete, unambiguous, verifiable, and necessary. Each requirement should be uniquely identified with a sequence number or a meaningful tag of some kind.»

REQ-1.1: «Some sentence goes here, in "shall" form, usually something like this: "Upon <some event or condition>, the <system, or the name of a module> shall <perform some action>." Note that the entire requirement must be stated in the requirements area. Everything else is just a lead-up to this. There should be one such entry per S/R pair.»

REQ-1.«N»: «Additional S/R-related requirements as needed.»

REQ-1.«N»: «List any additional requirements needed in “Shall” form, but not necessarily in “Upon/Shall” form.»

REQ-1.«N»: «Additional requirements as needed.»

5.«N». «Additional feature blocks»

6. Nonfunctional Requirements

6.1. Performance

NF-1.1: «Any performance-related issue»

NF-1.«N»: «Any performance-related issue»

6.2. Security

NF-2.1: «Any security-related issue»

NF-2.«N»: «Any security-related issue»

6.«N». «Any additional “nonfunctional” blocks»